

Data Validation

2026-04-09

Libraries Imported

```
library(ggplot2)
library(dplyr)
library(tidyverse)
library(forecast)
library(strucchange)
library(corrplot)
library(psych)
library(moments)
library(e1071)
library(tseries)
library(car)
```

Importing and Cleaning Data

When Reading the Data, I recognized that the GDP Nominal and GDP_PPP Metrics were identical. To resolve this issue, I removed 2 of the duplicate fields and renamed the other 2 fields as “GDP” and “GDP_Per_Capita”

```
original_df = read.csv("Economy_of_US.csv")

df <- original_df[, c(1, 2, 3, 6, 7, 8, 9)]
colnames(df)[2] <- "GDP"
colnames(df)[3] <- "GDP_PerCapita"

gdp_df <- df[, 2:3]
quant_vars <- df[, 2:6]
```

Descriptive Statistics

```
summary(quant_vars)
```

##	GDP	GDP_PerCapita	GDP_Growth	Inflation
## Min.	: 2857	Min. :12553	Min. : -0.03400	Min. : -0.00300
## 1st Qu.:	6430	1st Qu.:25120	1st Qu.: 0.01700	1st Qu.: 0.02000
## Median :	11837	Median :40524	Median : 0.02600	Median : 0.02800
## Mean :	13182	Mean :43192	Mean : 0.02446	Mean : 0.03221
## 3rd Qu.:	18328	3rd Qu.:57007	3rd Qu.: 0.03725	3rd Qu.: 0.03525

```
## Max.      :30282   Max.      :89546   Max.      : 0.07200   Max.      : 0.13500
## Unemployment
## Min.      :0.0370
## 1st Qu.:0.0485
## Median :0.0555
## Mean     :0.0605
## 3rd Qu.:0.0720
## Max.     :0.0970
```

```
# Returns a named vector of variances
sapply(quant_vars, var, na.rm = TRUE)
```

```
##           GDP GDP_PerCapita   GDP_Growth   Inflation Unemployment
## 6.111153e+07 4.578268e+08 3.845940e-04 5.603812e-04 2.686809e-04
```

```
#Analyze Skewness of Quantitative Variables
skew(quant_vars)
```

```
## [1] 0.5414018 0.4644225 -0.6995630 2.3386895 0.7057662
```

```
#Kurtosis
apply(quant_vars, 2, kurtosis, na.rm = TRUE)
```

```
##           GDP GDP_PerCapita   GDP_Growth   Inflation Unemployment
## -0.8180309 -0.8198000 1.2801250 6.8463908 -0.4494227
```

For Skewness, the metrics indicate the following:

- Highly skewed if the absolute value is greater than 1
- Moderately skewed if the absolute value is between 0.5 and 1
- Symmetric if the absolute value is less or equal to 0.5

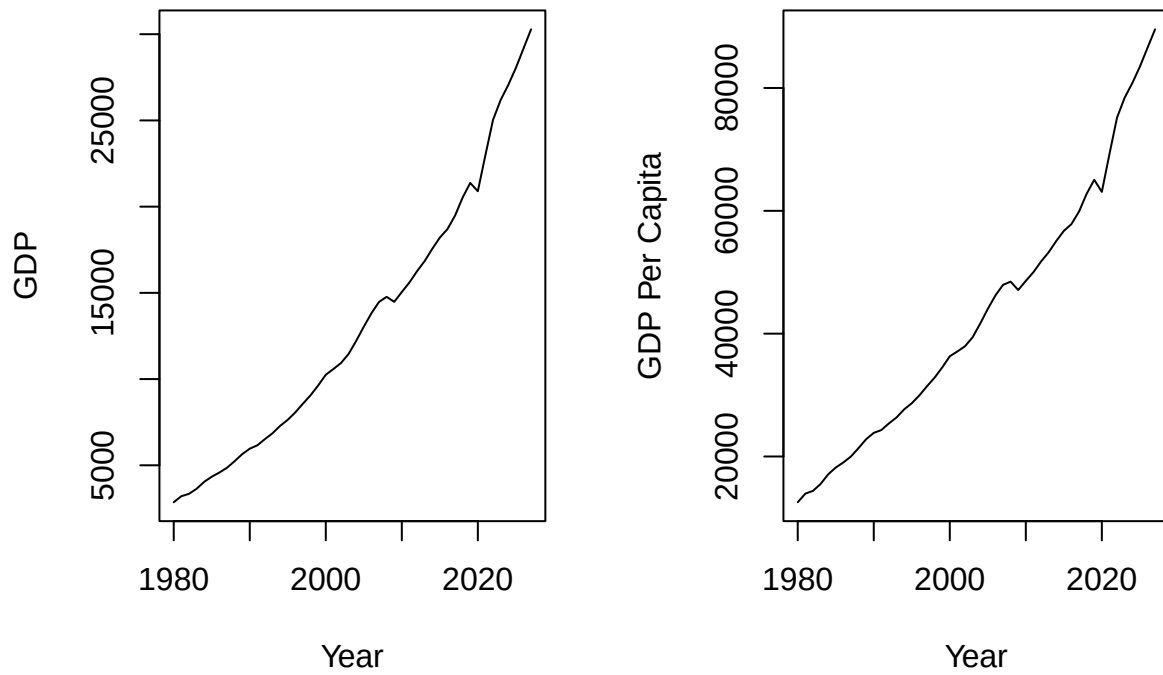
From the results, the each metric is distributed as follows:

- Unemployment is skewed
- GDP and GDP Per Capita is moderately skewed
- GDP Growth is Highly Skewed

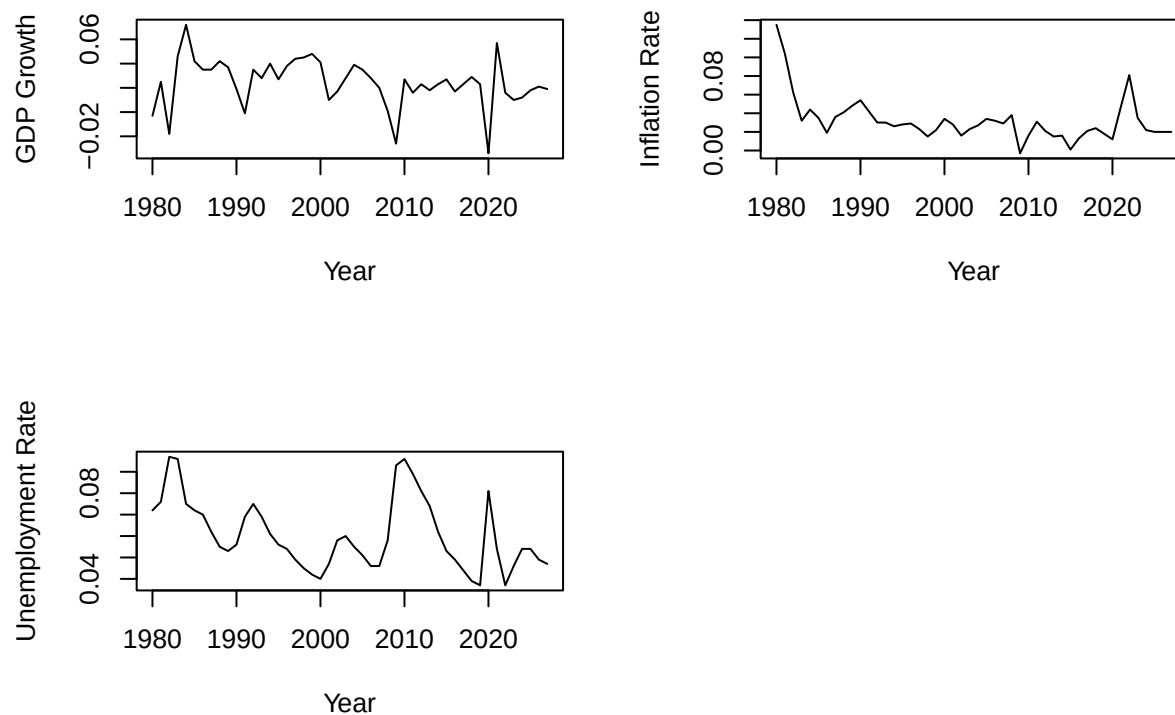
Distribution Plots

```
par(mfrow = c(1, 2))

plot(df$Year, df$GDP, type = "l", xlab="Year", ylab = "GDP")
plot(df$Year, df$GDP_PerCapita, type = "l", , xlab="Year", ylab = "GDP Per Capita")
```



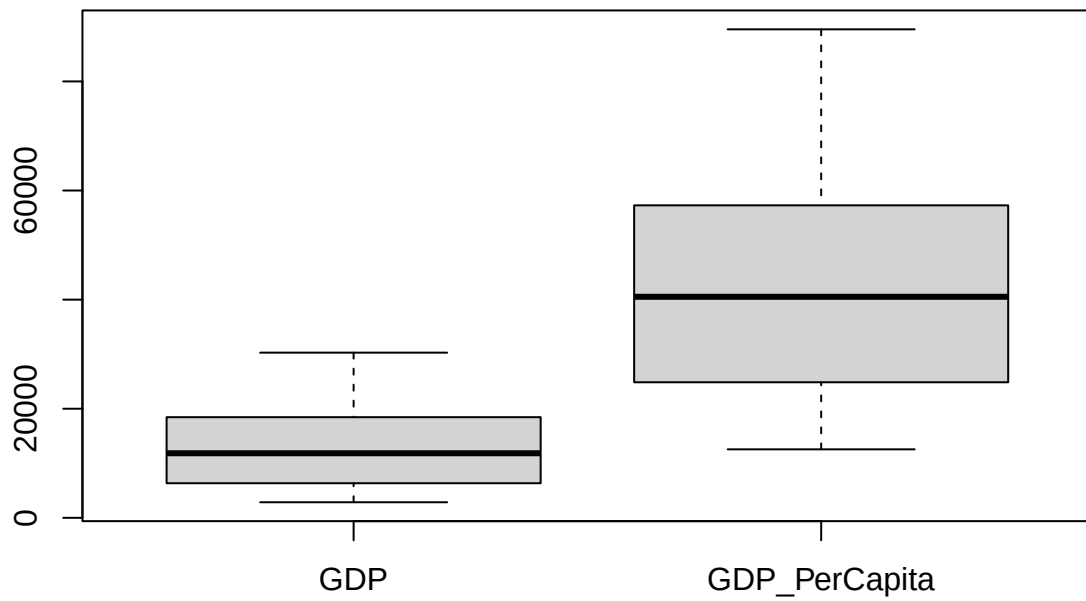
```
par(mfrow = c(2, 2))
plot(df$Year, df$GDP_Growth, type = "l", xlab="Year", ylab = "GDP Growth")
plot(df$Year, df$Inflation, type = "l", xlab="Year", ylab = "Inflation Rate")
plot(df$Year, df$Unemployment, type = "l", xlab="Year", ylab = "Unemployment Rate")
```



To identify anonymous outliers in the data, I used a box plot for each quantitative metric to identify any possible outliers. Here, there are some in growth and unemployment, but because there is a relatively low number of records initially in the data, I decided not to transform them, but considered them for the modelling process.

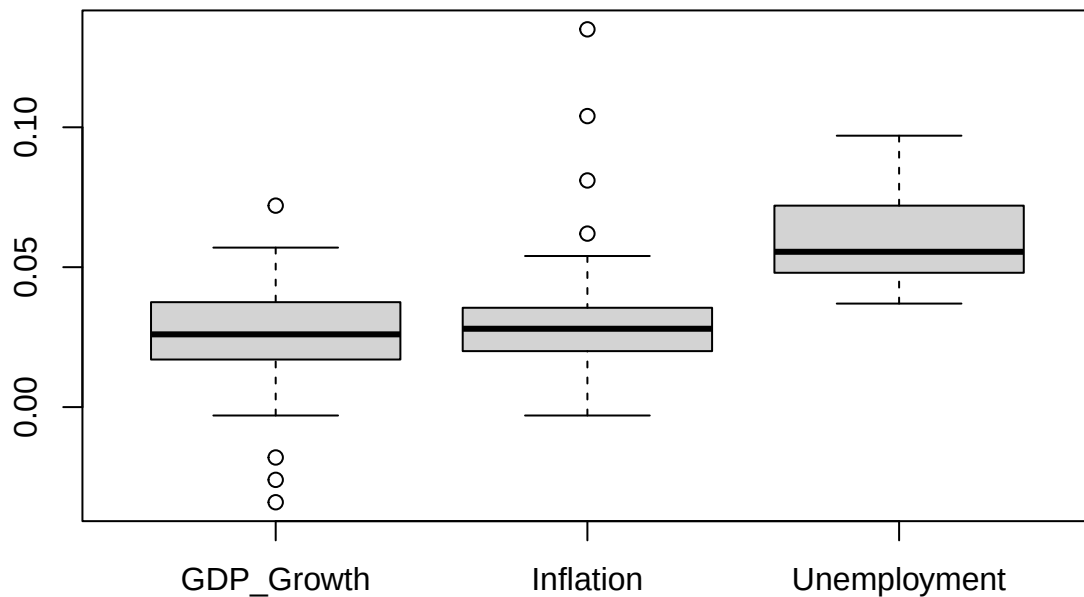
```
boxplot(quant_vars[,1:2], main = "Box Plot Distributions for Quantitative Variables")
```

Box Plot Distributions for Quantitative Variables



```
boxplot(quant_vars[,3:5], main = "Box Plot Distributions for GDP Metrics and Factors")
```

Box Plot Distributions for GDP Metrics and Factors



Hypothesis Testing

ANOVA

First, I added a column to label years based on decades. Here, I classified the years are based on the shared “Tens Digit”, and assigned a decade as the starting year it is part of.

```
df$decade <- as.character(floor(df$Year/10)*10)
```

```
anova_model <- aov(GDP ~ decade, data = df)
summary(anova_model)
```

```
##           Df    Sum Sq   Mean Sq F value Pr(>F)
## decade      4 2.714e+09 678619883    185 <2e-16 ***
## Residuals   43 1.578e+08   3668889
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The most important result to determine the Hypothesis is the P Value. Here, there is a Null Hypothesis, which states there is a difference between at least 2 decade GDP averages. If there is a 5% allowed chance of error in the data, then because the p-value is significantly less than 0.05, the null hypothesis is rejected and ANOVA concludes that the average GDP across all decades is not equal

Pearson's R

```
cor(x=df$GDP, y = df$GDP_PerCapita, method = "pearson")
```

```
## [1] 0.9991232
```

```
cor(x=df$GDP_PerCapita, y = df$GDP_Growth, method = "pearson")
```

```
## [1] -0.2476102
```

```
cor(x=df$GDP, y = df$GDP_Growth, method = "pearson")
```

```
## [1] -0.2560569
```

From this test, the following can be observed: - There is a nearly perfect positive correlation between GDP and GDP Per Capita - There is a weak relation between GDP Growth and each with GDP and GDP Per Capita

```
cor.test(x=df$GDP, y = df$GDP_PerCapita, method = "pearson")
```

```
##
## Pearson's product-moment correlation
##
## data: df$GDP and df$GDP_PerCapita
## t = 161.85, df = 46, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.9984277 0.9995111
## sample estimates:
## cor
## 0.9991232
```

```
cor.test(x=df$GDP_PerCapita, y = df$GDP_Growth, method = "pearson")
```

```
##
## Pearson's product-moment correlation
##
## data: df$GDP_PerCapita and df$GDP_Growth
## t = -1.7334, df = 46, p-value = 0.08973
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.49679318 0.03928865
## sample estimates:
## cor
## -0.2476102
```

```
cor.test(x=df$GDP, y = df$GDP_Growth, method = "pearson")
```

```
##
## Pearson's product-moment correlation
##
## data: df$GDP and df$GDP_Growth
## t = -1.7966, df = 46, p-value = 0.07897
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.50355555 0.03028093
## sample estimates:
## cor
## -0.2560569
```

Across all pairs of GDP Metrics, when willing to give 5% error, since all p values are greater than 0.05, reject the null hypothesis among each tested pair and conclude each correlation is statistically significant.

Augmented Dickey Fuller Test

```
apply(quant_vars, 2, adf.test)
```

```
## Warning in FUN(newX[, i], ...): p-value greater than printed p-value
## Warning in FUN(newX[, i], ...): p-value greater than printed p-value
## Warning in FUN(newX[, i], ...): p-value smaller than printed p-value

## $GDP
##
## Augmented Dickey-Fuller Test
##
## data: newX[, i]
## Dickey-Fuller = 1.2556, Lag order = 3, p-value = 0.99
## alternative hypothesis: stationary
##
##
## $GDP_PerCapita
##
## Augmented Dickey-Fuller Test
##
## data: newX[, i]
## Dickey-Fuller = 1.0115, Lag order = 3, p-value = 0.99
## alternative hypothesis: stationary
##
##
## $GDP_Growth
##
## Augmented Dickey-Fuller Test
##
## data: newX[, i]
## Dickey-Fuller = -4.2558, Lag order = 3, p-value = 0.01
## alternative hypothesis: stationary
```



```
##
##
## $Inflation
##
## Augmented Dickey-Fuller Test
##
## data: newX[, i]
## Dickey-Fuller = -3.2323, Lag order = 3, p-value = 0.09306
## alternative hypothesis: stationary
##
##
## $Unemployment
##
## Augmented Dickey-Fuller Test
##
## data: newX[, i]
## Dickey-Fuller = -3.1577, Lag order = 3, p-value = 0.1118
## alternative hypothesis: stationary
```

The Augmented Dickey Fuller Test determines if the time series data is has an identifiable trend or not. From the results, if a 5% error is allowed in the data, the null hypothesis is rejected for Growth Rates. Here, it can be concluded that the Growth Rate over time is stationary. From the previous line graph plotted, there is not a recognizable trend.

Correlation Analysis

```
corrplot(cor(quant_vars), type = "upper", order = "hclust",
          tl.col = "black", tl.srt = 45)
```



Variance Inflation Factor

Here, the model to evaluate is between Growth Rate and Inflation and Unemployment

```
model <- lm(GDP_Growth ~ Inflation + Unemployment, data = quant_vars)
summary(model)
```

```
##
## Call:
## lm(formula = GDP_Growth ~ Inflation + Unemployment, data = quant_vars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.052009 -0.009714  0.000208  0.012823  0.052596
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.04523    0.01111   4.071 0.000187 ***
## Inflation    -0.01886    0.11896  -0.159 0.874733
## Unemployment -0.33322    0.17180  -1.939 0.058725 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01923 on 45 degrees of freedom
## Multiple R-squared:  0.0792, Adjusted R-squared:  0.03827
## F-statistic: 1.935 on 2 and 45 DF, p-value: 0.1562
```

```
vif(model)
```

```
##      Inflation Unemployment  
##      1.007747      1.007747
```

Here, there is a moderate correlation between each Inflation and Unemployment with GDP Growth Rates. With this, no variable reduction is necessary, but this could affect model performance and accuracy, in which it will be considered.